StudentEnrollment&CourseManagementSystem(secms)

ProjectworksubmittedtoThiruvalluvarUniversity,Serkkadu, Vellore.*In*

partial fulfillment for the award of degree

BACHELOROFCOMPUTERAPPLICATION

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DEPARTMENT OFCOMPUTER APPLICATION

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MARCH– 2026

StudentEnrollment&CourseManagementSystem(SECMS)

ProjectDocumentation

1. Introduction

1.1 ProjectOverview

The**StudentEnrollment&CourseManagementSystem(SECMS)**isaSalesforce-based application developed to automate and centralize academic enrollment activities in an educational institution. The system manages student records, courses, instructors, enrollments, approvals, payments, reports, and dashboards within a single platform.

1.2 PurposeoftheProject

ThepurposeofSECMSistoreplace manualand fragmentedenrollment processeswithan automated, scalable, and secure Salesforce solution that ensures:

- Accuratestudentdatamanagement
- Structuredenrollmentworkflows
- Automatedvalidationandapprovalprocesses
- Real-timereportingandmonitoring

2. SystemObjectives

- CentralizeallacademicdatainSalesforce
- Reducemanualeffortandhumanerrors
- Improvetransparencyinenrollmentandrevenuetracking
- Providerole-basedaccessandsecurity
- Enabledata-drivendecisionmakingusingreportsanddashboards

3. TechnologyStack

Layer	TechnologyUsed
UserInterface	SalesforceLightningExperience
BusinessLogic	SalesforceFlows
Database	SalesforceCustomObjects
Automation	ValidationRules,ApprovalProcesses
Reporting	SalesforceReports&Dashboards
Security	Profiles,PermissionSets,OWD

4. SystemArchitectureOverview

The SECMS architecture consists of Salesforce custom objects (Student, Course, Instructor, Enrollment)integratedwithautomationtoolssuchasFlowsand ApprovalProcesses. Reports and Dashboards consume processed data for real-time monitoring, while securityis enforced using profiles and permission sets.

4.1 ArchitecturalLayers

1. PresentationLayer(UserInterface)
Thislayerrepresentstheuser-facingcomponentsofthesystemand isbuilt using**Salesforce Lightning Experience**. It provides an intuitive and responsive interface for all users, including Administrators, Enrollment Officers, and Instructors.

- LightningAppsandTabs fornavigation
- RecordpagesforStudent,Course,Instructor,andEnrollmentobjects
- DashboardsandReportsforreal-timeinsights

2. ApplicationLayer(BusinessLogic&Automation)
The application layer handles all business rules and automation logic using Salesforce declarativetools.Thislayerensuresthat

enrollmentworkflowsareexecutedaccuratelyandconsistently.

- SalesforceFlowsforautomation(record-triggeredflows)
- ValidationRules forenforcingdataintegrity
- ApprovalProcessesforenrollmentauthorization
- RecordTypesandPageLayoutsfor processdifferentiation

3. DataLayer(Database)

ThedatalayerconsistsofSalesforcecustomobjectsthatstoreand manageacademic information securely.

- Studentobjectstorespersonaland academicstatusdetails
- Courseobjectmanagescourseofferingsandfees
- Instructorobjectmaintainsfacultyinformation
- Enrollmentobjectactsasajunctionobjectlinkingstudents,courses,andinstructors

RelationshipsbetweenobjectsareestablishedusingLookuprelationshipsto ensure referential integrity and structured data access.

4. SecurityLayer
Securityisenforcedacrossalllayersusing Salesforce' sbuilt-insecurityframework.

- ProfilesdefinebaselineaccessforAdmin,EnrollmentOfficer,andInstructorroles
- PermissionSetsprovideadditional,flexibleaccesscontrol
- Organization-WideDefaults(OWD) controlrecord-levelvisibility
- Field-LevelSecurityensuressensitivedataisprotected

5. Reporting&AnalyticsLayer
Thislayertransformsrawdataintomeaningfulinsightsfordecision-making.

- StandardandcustomReportsforenrollment,revenue, andapprovals
- InteractiveDashboardsfor visualmonitoringofKPIs
- Real-timedataupdatesreflectingsystemactivities

4.2 SystemWorkflowOverview

1. Usersaccessthesystemthroughthe SalesforceLightninginterface.
2. Student, Course,andEnrollmentdataisenteredthroughstructuredrecordpages.
3. ValidationRulesverifydataaccuracybeforerecordsubmission.
4. Record-TriggeredFlowsautomateenrollmentdatepopulation,instructorassignment, and task creation.
5. EnrollmentrequestsareroutedthroughApprovalProcesses.
6. Approvedenrollmentstriggernotificationsandstatusupdates.
7. ReportsandDashboardsconsumeprocesseddataformonitoringandanalysis.

4.3 ArchitectureBenefits

- **Scalability:**Easilysupportsincreasingstudents, courses,andusers
- **Reliability:**Salesforcecloudensuresconsistentavailability
- **Security:**Strongrole-basedaccessanddataprotection
- **Maintainability:**Declarativeconfigurationreducesmaintenanceeffort
- **Flexibility:**Easyintegrationwithfuturemodulessuchaspayment gatewaysandportals

5. DataModelandRecords

5.1 StudentObject

FieldName	DataType	Description
StudentName	Text	Nameofthestudent
Email	Email	InstitutionalemailID
Phone	Phone	Contactnumber
DateofBirth	Date	StudentDOB
StudentStatus	Picklist	Active/Inactive
Address	TextArea	Residentialaddress

SampleStudentRecord

StudentName	Email	Status
VIGNESH	ppvignesh516@gmail.com	Active

5.2 CourseObject

FieldName	DataType	Description
CourseName	Text	Nameofthecourse
Category	Picklist	Technical/Language/Non-Technical
Duration	Number	Courseduration
CourseFee	Currency	Feeamount
Description	TextArea	Coursedetails

SampleCourseRecord

CourseName	Category
DATASCIENCE	ADCEMICYEAR

5.3 InstructorObject

FieldName	DataType	Description
InstructorName	Text	Nameofinstructor
InstructorCode	Text	Uniquecode
Expertise	Picklist	Subjectexpertise
Phone	Phone	Contactnumber
Email	Email	EmailID

SampleInstructorRecord

InstructorName	Expertise
vignesh	Requested

5.4 EnrollmentObject

FieldName	DataType	Description
Student	Lookup(Student)	Enrolledstudent
Course	Lookup(Course)	Selectedcourse
Instructor	Lookup(Instructor)	Assignedinstructor
EnrollmentDate	Date	Dateofenrollment
EnrollmentStatus	Picklist	Requested/Approved/Rejected
FeesPaid	Checkbox	Paymentstatus
TotalAmount	Currency	Feeamount

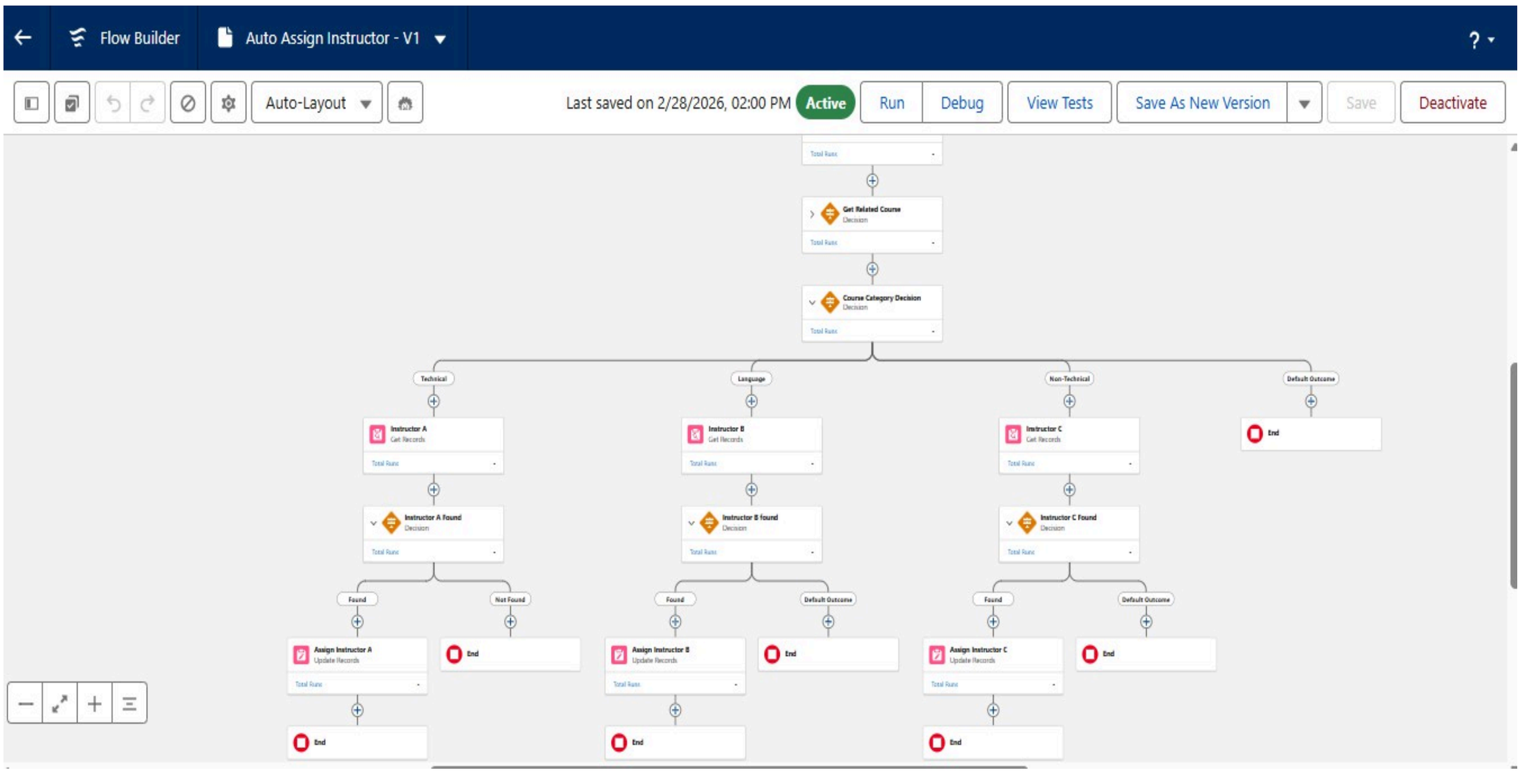
SampleEnrollmentRecord

Student	Course	Status	FeesPaid
Vignesh	Datascience	active	Yes

6. FunctionalModulesImplemented

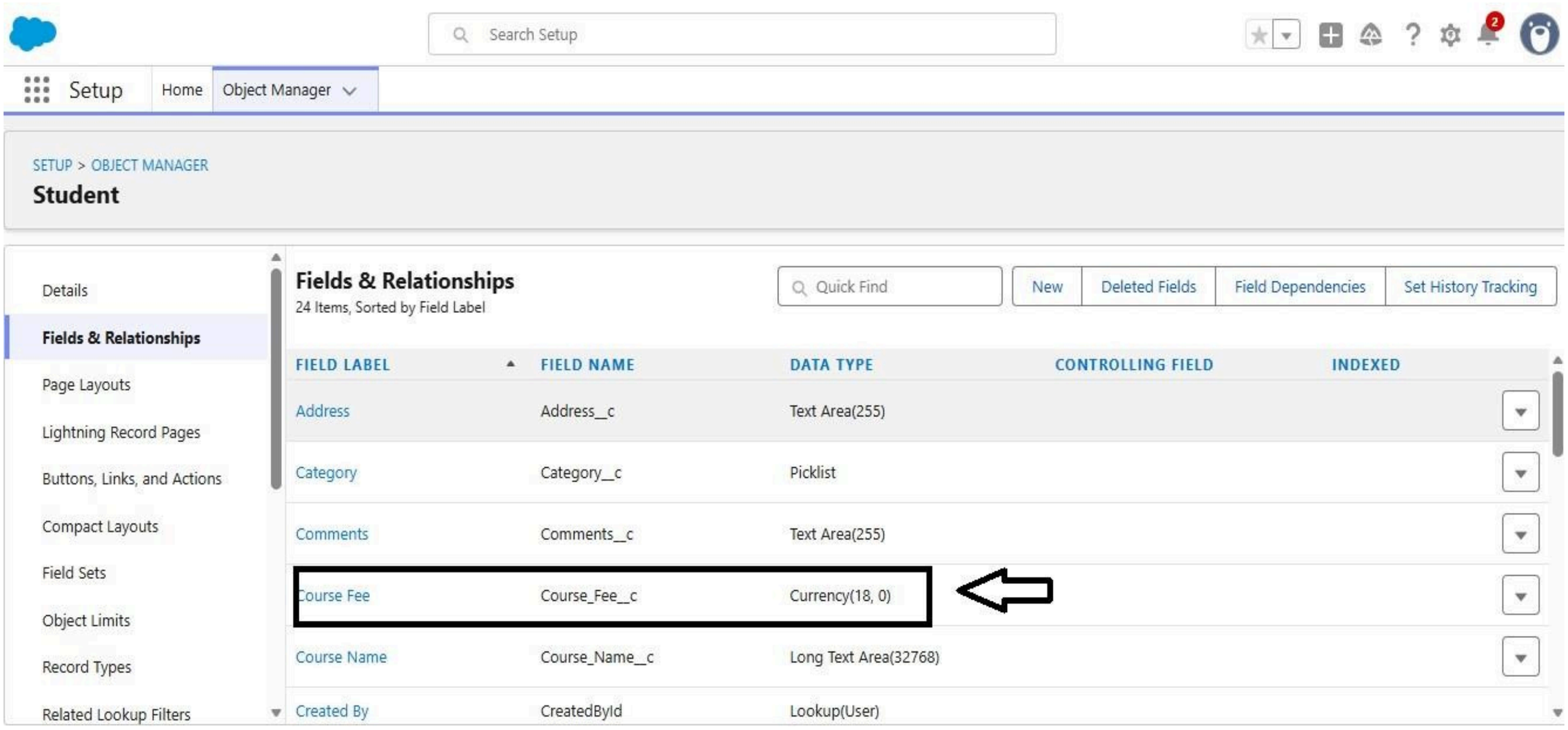
6.1 StudentManagement

- Createandmanagestudentrecords
- Emailvalidationusingvalidationrules



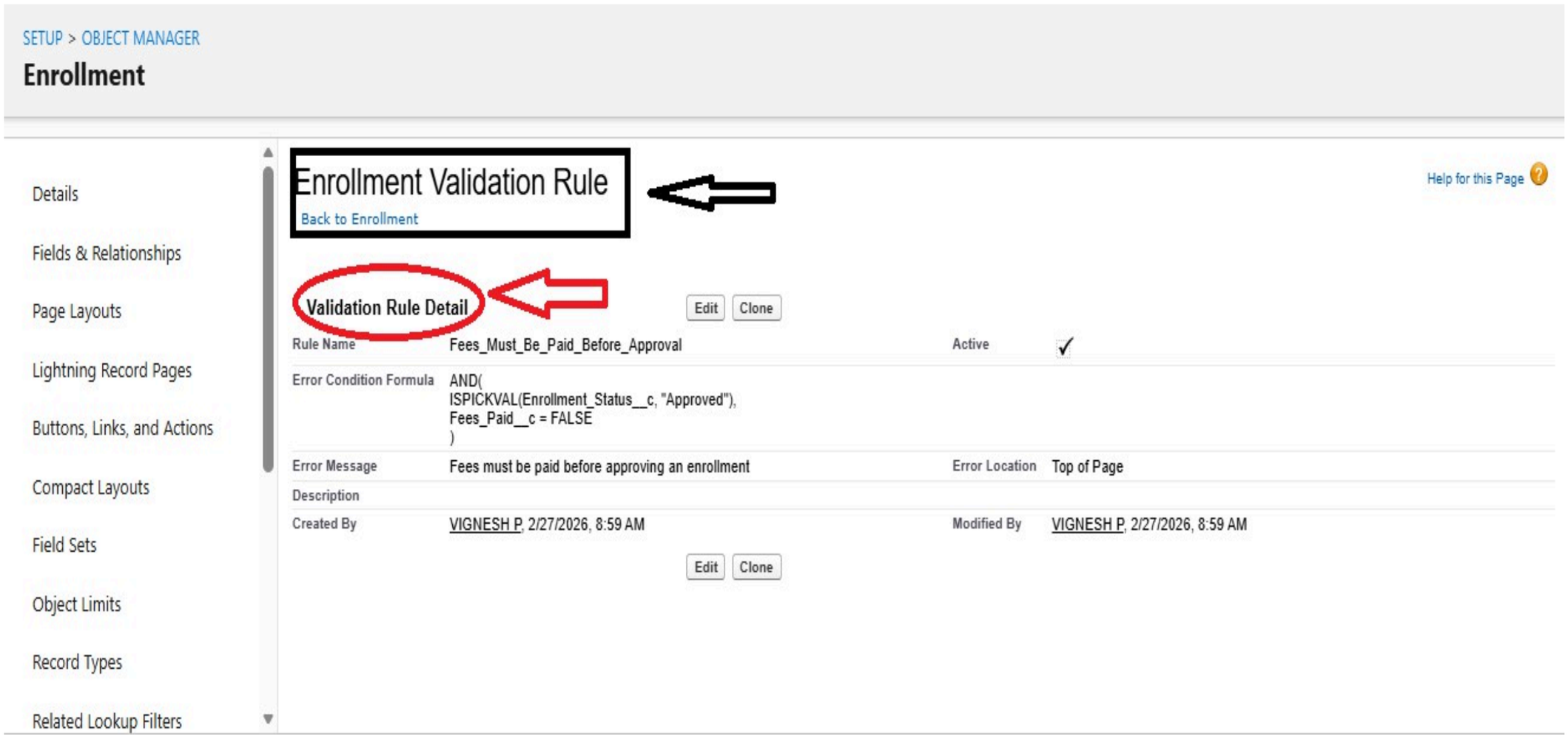
6.2 CourseManagement

- Course creationwithcategoryandfees
- Courseavailabilityforenrollments



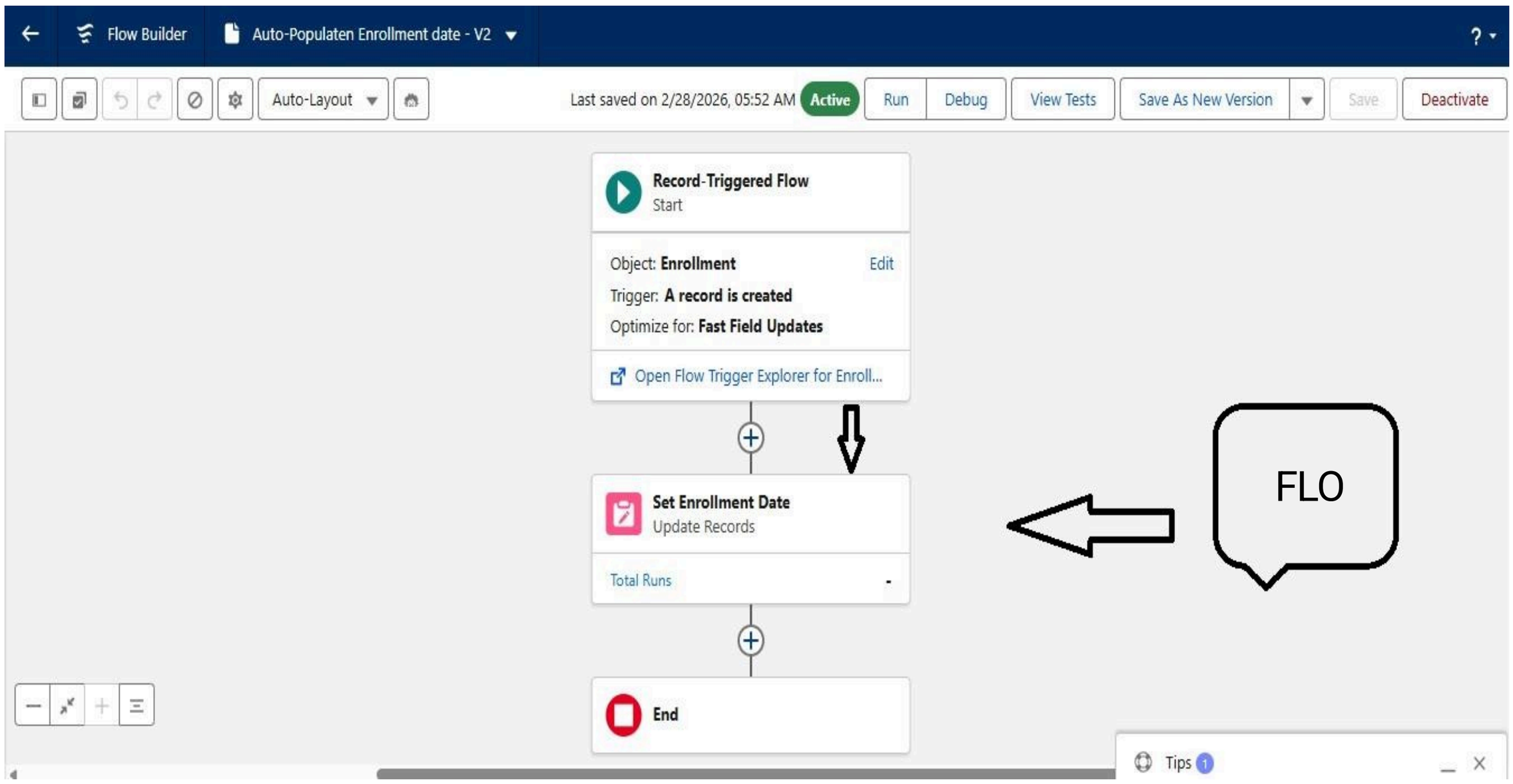
6.3 EnrollmentProcessing

- NewEnrollmentandRe-Enrollmentrecordtypes
- Validationforfeepaymentbeforeapproval



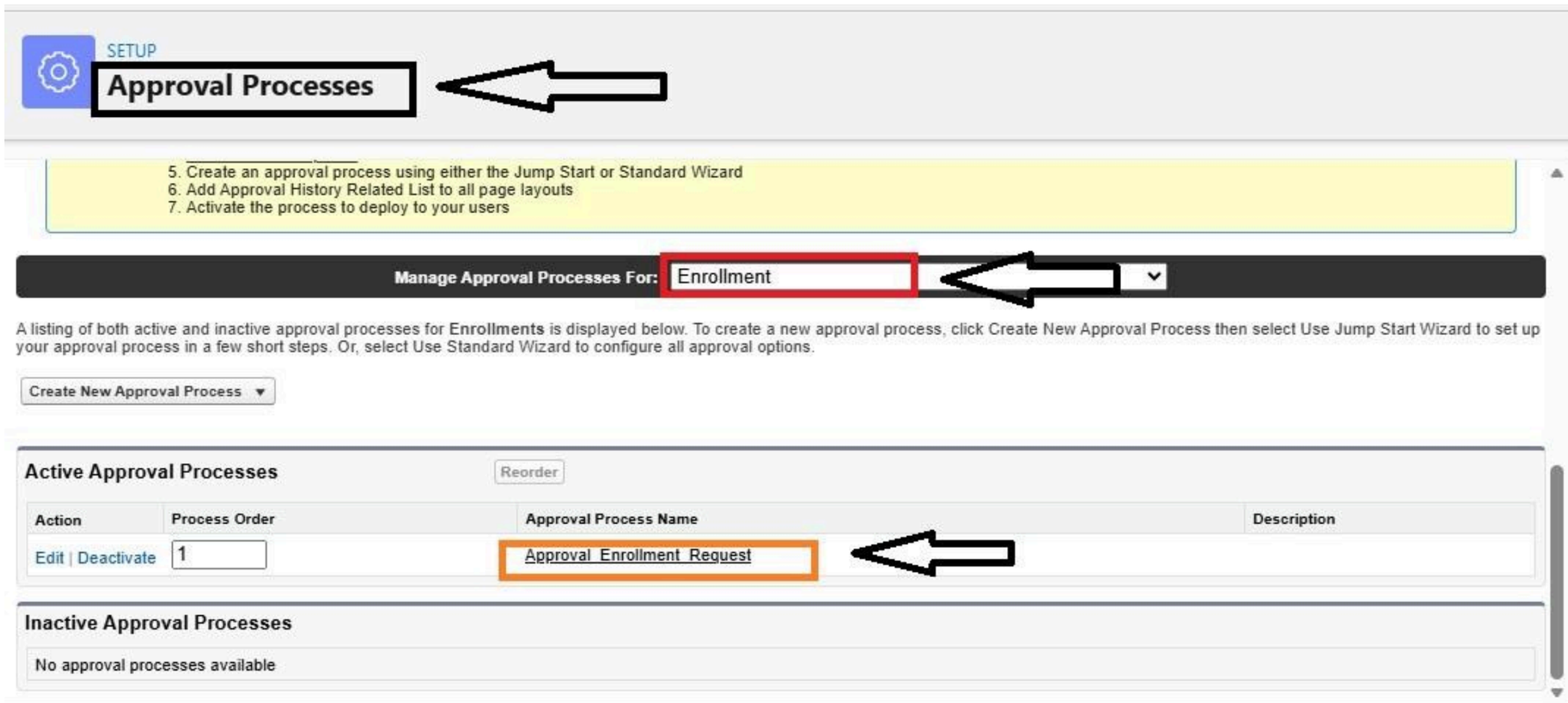
6.4 Automation(Flows)

- Auto-populateenrollmentdate
- Auto-assigninstructorsbasedoncoursecategory
- Auto-createfollow-uptasks
- Emailnotificationonapproval



6.5 ApprovalProcess

- EnrollmentrequestsroutedtoAdmin/Manager
- Autoupdatestatusonapprovalorrejection



7. ReportsandDashboa

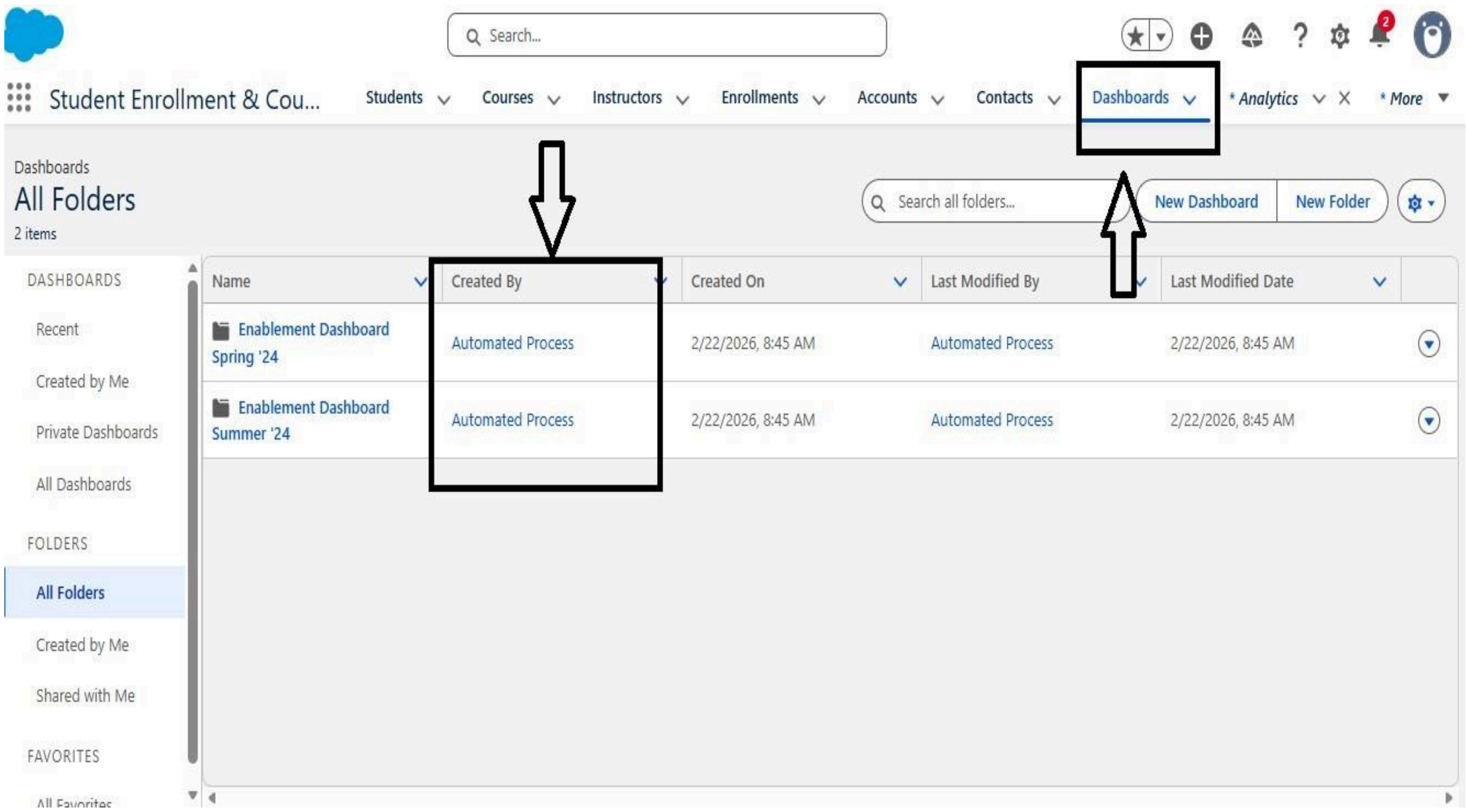
rds Reports Created

ReportName	Purpose
StudentsbyStatus	Trackactive/inactivestudents
EnrollmentsbyCourse	Monitorcoursedemand
PendingEnrollmentApprovals	Trackapprovals
RevenueReport	Trackfeescollected

Dashboard

StudentManagementDashboardincludes:

- Donutchartforstudentstatus
- Barchartforenrollmentsbycourse
- Gaugeforrevenue
- Tableforpendingapprovals



8. SecurityConfiguration

Role	AccessLevel
Admin	FullAccess
EnrollmentOfficer	Create/EditEnrollments
Instructor	Read-onlyassignedenrollments

Securityfeaturesimplemented:

- ProfilesandPermissionSets
- OrganizationWideDefaults(OWD)
- Role-basedaccesscontrol

9. TestingSummary

Comprehensive functionalandperformancetesting wascarriedoutto validatethatthe Student Enrollment&Course Management System (SECMS) meets all specified functional and non-functional requirements. Testing activities were conducted in the Salesforce Developer Environment to ensure system stability, accuracy, and reliability before finalsubmission.

9.1 FunctionalTesting

Functionaltesting focusedonverifyingthat eachmoduleofthesystemoperates according to the defined requirements.

StudentManagementTesting

- Verifiedcreation,editing,anddeletionofStudentrecords
- Validatedmandatoryfieldchecks(Name,Email,Status)
- Testedinstitutionalemaildomainvalidationrule

CourseManagementTesting

- ConfirmedsuccessfulcreationandupdateofCourserecords
- Verifiedcorrectdisplayofcoursecategory, duration, andfee
- Ensuredcoursesareselectableduringenrollment

EnrollmentProcessingTesting

- TestedNewEnrollmentandRe-Enrollmentrecordtypes
- VerifiedlookuprelationshipsbetweenStudent,Course, Instructor, andEnrollment
- EnsuredEnrollmentStatusupdatescorrectly(Requested,Approved,Rejected)

ValidationRuleTesting

- Confirmederrormessagesappearwhenfeesarenotpaidbeforeapproval
- Verifiedpreventionofinvalidemailformats

ApprovalProcessTesting

- Testedsubmissionofenrollmentrequestsforapproval
- Verifiedroutingtothecorrectapprover
- Confirmedautomaticstatusupdatesonapprovalandrejection
- Testedrejectionemailnotifications

Automation(Flow)Testing

- VerifiedauteopulationofEnrollmentDateonrecordcreation
- Testedautomaticinstructorassignmentbasedoncoursecategory
- Verifiedautomatictaskcreationforpendingenrollments
- Confirmedemailnotificationsonenrollmentapproval

9.2 PerformanceTesting

Performancetestingensuredthesystemperformsefficientlyunder normalandpeakusageconditions.

- Verifiedquickrecordcreationandupdateresponsetimes
- Testedbulkrecordinsertionwithoutsystemerrors
- Ensureddashboardsloadwithout noticeabledelay
- Verifiedreportsgenerateaccurateresultswithinacceptabletime

9.3 SecurityTesting

Securitytestingvalidatedthatdataaccess isrestrictedbasedonuserroles.

- VerifiedAdminprofilehas fullaccess
- ConfirmedEnrollmentOfficerscancreateandupdateenrollmentsonly
- VerifiedInstructorscanviewonlyassignedenrollments
- TestedOrganization-WideDefaults(OWD)behavior

9.4 ReportingandDashboardValidation

- Verifiedaccuracyofdatadisplayedinreports
- Testedgroupingandfilteringlogicinreports
- Confirmeddashboardsreflectreal-timedataupdates

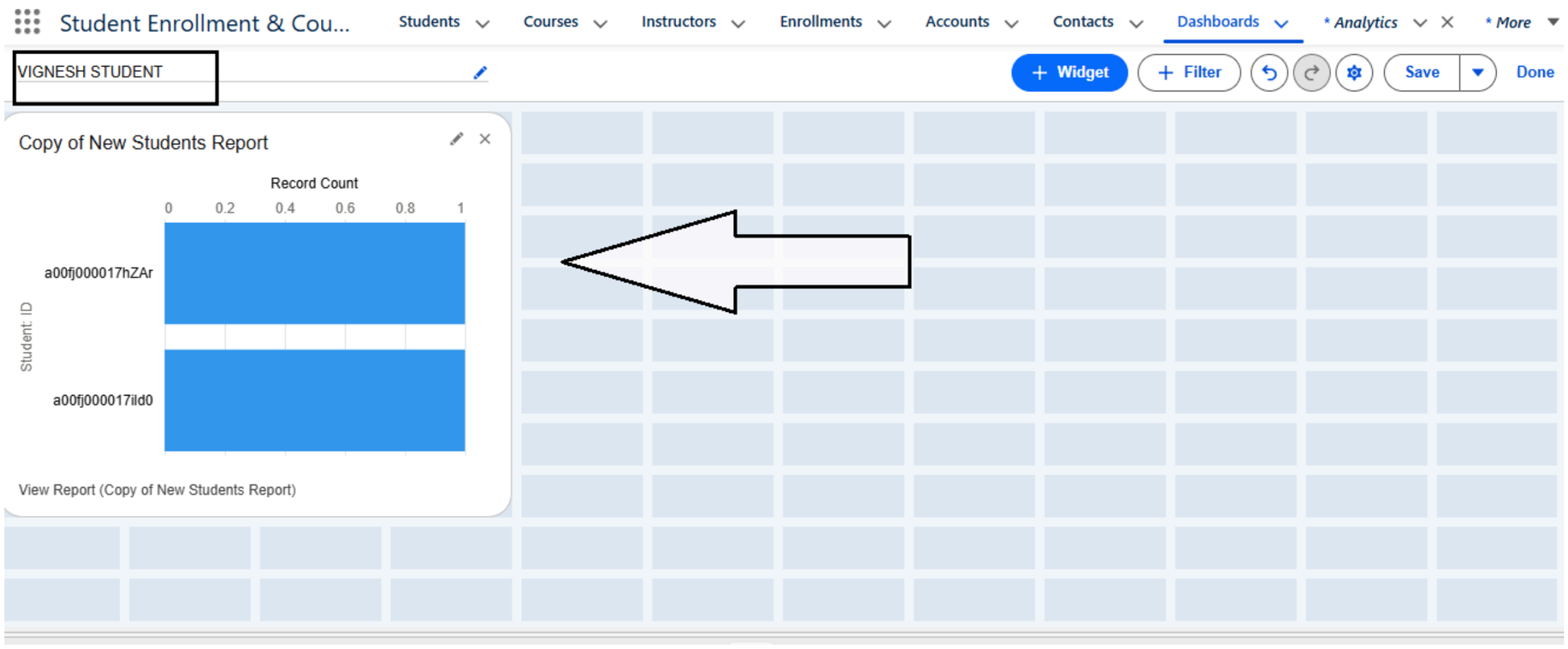
9.5 TestEvidence

Screenshots were captured for all major testing activities, including record creation, validation errors, approval actions, automated flow executions, reports, and dashboards.
Thesescreenshotsserveasdocumentedproofofsuccessfulsystemtestingand functionalityverification.

Functionalandperformancetesting wasconductedtovalidate:

- Recordcreationandupdates
- Validationrulesenforcement
- Approvalworkflows
- Automationflowsexecution
- Accuracyofreportsanddashboards

Screenshotswerecapturedasevidenceduringtesting.



10. AdvantagesandDisadvantag

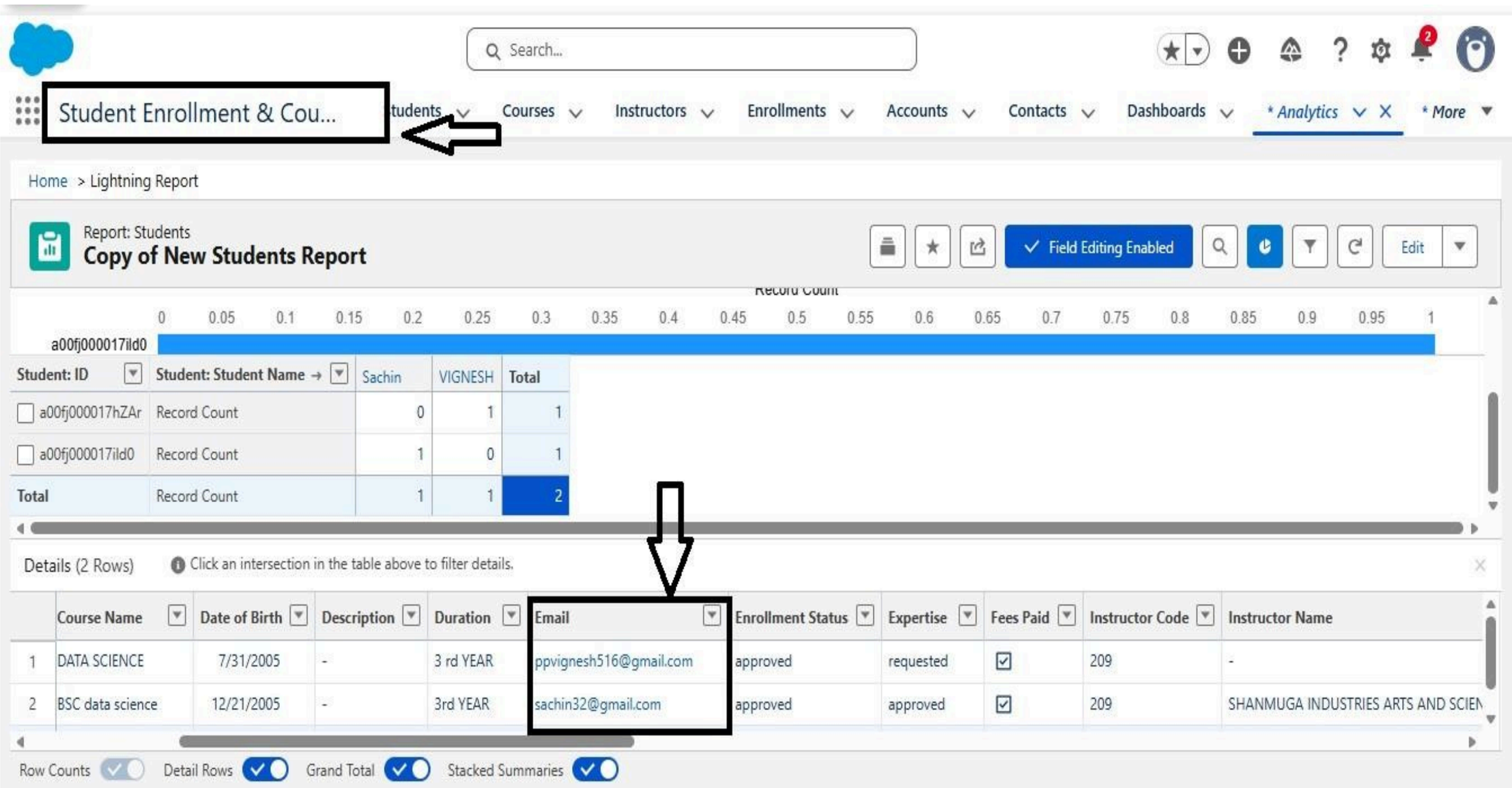
esAdvantages

- Centralizeddatamanagement
- Reducedmanualwork
- Real-timeinsights
- Scalablearchitecture

Disadvantages

- DependencyonSalesforceplatform
- Initialsetup andconfigurationeffort

Resultoutput:


Search...
Student Enrollment & Cou...
Students
Courses
Instructors
Enrollments
Accounts
Contacts
Dashboards
* Analytics
* More
Home > Lightning Report
Report: Students
Copy of New Students Report
Field Editing Enabled
Edit
Record Count
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9 0.95 1
a00fj000017ild0
Student: ID | Student: Student Name | Sachin | VIGNESH | Total
a00fj000017hZAr | Record Count | 0 | 1 | 1
a00fj000017ild0 | Record Count | 1 | 0 | 1
Total | Record Count | 1 | 1 | 2
Details (2 Rows)
Click an intersection in the table above to filter details.
Course Name | Date of Birth | Description | Duration | Email | Enrollment Status | Expertise | Fees Paid | Instructor Code | Instructor Name
1 | DATA SCIENCE | 7/31/2005 | - | 3 rd YEAR | ppvignesh516@gmail.com | approved | requested | 209 | -
2 | BSC data science | 12/21/2005 | - | 3rd YEAR | sachin32@gmail.com | approved | approved | 209 | SHANMUGA INDUSTRIES ARTS AND SCIEN
Row Counts
Detail Rows
Grand Total
Stacked Summaries

11. Conclusion

The Student Enrollment&Course Management System (SECMS) has been successfullydesigned, developed, and implemented using the Salesforce platformto address the challenges of manual, fragmented, and error-prone academic enrollment processes. This project delivers a complete end-to-end solution that manages the entire student lifecycle— fromstudent registrationandcoursecreationtoenrollment processing, instructorassignment, approvals, fee tracking, and performance reporting.

By leveraging Salesforce core features such as Custom Objects, Validation Rules, Record Types, Approval Processes, and Flows, the system ensures data accuracy, consistency, and automation across all academic operations. Manual intervention has been significantlyreduced,whichminimizeserrors,avoidsduplicateenrollments,andaccelerates decision-making processes. Automated validations and approvals ensure that institutional policies— such as fee payment before approval— are strictly enforced.

The implementation of role-based security using Profiles, Permission Sets, and Organization-Wide Defaults (OWD) ensures that sensitive student and academic data is securely accessed only by authorized users. Administrators have complete visibility and control,enrollmentofficerscanefficiently manageregistrations,andinstructorscanview only their assigned enrollments, maintaining confidentiality and data integrity.

The system' s Reports and Dashboards provide real-time insights into key performance indicators such as student status distribution, course-wise enrollment trends, pending approvals, and revenue generated through fees. These analytics empower management tomakeinformed,data-drivendecisionsandproactivelyrespondtoenrollment patterns and operational bottlenecks.

Overall, SECMS demonstrates how Salesforce can be effectively used as a powerful academic management platform. The project not only fulfills all functional and non- functional requirements but also offers scalability, reliability, and flexibility for future enhancements. It serves as a strong foundation for digital transformation in educational institutionsandhighlightsbest practicesincloud-basedapplicationdevelopment,automation, and analytics.

12. FutureScope

- Paymentgatewayintegration
- Studentself-serviceportal

- Mobileapplicationsupport
- AdvancedanalyticsandAIinsights

13. Appendix

- SalesforceConfigurationDetails
- CustomObjectsandFields
- ReportsandDashboards

Projectanalysislink:

https://orgfarm-1d8fcddb6a-dev-ed.develop.my.salesforce.com/analytics/wave/report?assetId=00Ofj000008Z0VBEA0&orgId=00Dfj00000L7gsr&loginHost=usa1044.sfdc-8tgtt5.salesforce.com&urlType=sharing&analyticsContext=analyticsTab

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